

Cyber defence HQ

Revise 360 lesson plan — 60 minutes

Unit	OCR J277 2.3 Producing robust programs — Paper 2, Computational thinking, algorithms and programming
Lesson	Cyber defence HQ (lesson 2 of the unit)
Lesson ID	rp-l02
Experience ID	rp-l02
Plan ID	PLAN-RP-L02
Content version	697444c4b3c2
Website	https://revise360.co.uk/experience.html?id=rp-l02

Big question, outcomes and success criteria

Big question — How can a program defend itself against careless users and deliberate attackers?

Prerequisites

The lesson before this one is rp-l01. Pupils should already have finished it: the starter below retrieves from it.

Assessable outcomes

1. **rp-l02-o1** — know what is meant by defensive design considerations
2. **rp-l02-o2** — know a range of problems that can occur when a program is running
3. **rp-l02-o3** — understand some authentication techniques programmers use

Vocabulary, resources and preparation

Key vocabulary — Authentication, Input validation

Resources

SLIDES	presentations/RP_L02_CyberDefenceHQ_Lesson.pptx
WORKSHEET	worksheets/RP_L02_CyberDefenceHQ_Worksheet.docx
ONLINE EXPERIENCE	experience.html?id=rp-l02

Before the lesson

1. Sign in on the **teacher dashboard** (teacher.html) and open **Class logins**. Enter the class or group and the usernames, one per line, then print the login cards. A pupil cannot sign in without one, so do this before the lesson rather than in it.
2. Print the worksheet, one per pupil.
3. Open the slides and the experience once yourself, on the machine you will be teaching from, so the 360 image is already cached.
4. Check headsets or devices are charged and on the school network.

How pupils get in

1. Go to `topics.html` on `revise360.co.uk`.
2. Sign in with the **username and PIN from the login card**. The class is already set against the card.

3. Choose the topic, then this lesson's card.

Worth knowing before you promise anything. In this build APP_CONFIG.backendUrl is empty, so logins, login cards and progress are stored on the device the pupil used. The teacher dashboard says so too. A pupil who moves to a different machine starts again, so keep a class on the same devices within a unit, and treat the paper worksheet as the record that travels.

Teaching sequence

Sixty minutes including transitions. The online block is the one to move if your lesson is shorter: see the adaptation at the end.

MIN	PHASE	TEACHER	PUPILS	EVIDENCE TO CHECK
0-5	Retrieve	Starter on the board or the worksheet, from the last lesson and this lesson's prerequisites.	Answer the starter on the worksheet.	The starter box on RP_L02_CyberDefenceHQ_Worksheet.docx is filled in.
5-8	Review the starter	Take answers; correct the one most got wrong.	Correct their own starter.	A correction written, not just a tick.
8-18	Explain and model	The new idea in two or three steps, with the example worked in front of the class.	Listen and answer questions about the example.	They can say why, not only what.
18-21	Set up and settle	Sign-in, launch, and what to write before anyone taps an answer.	Sign in, open the lesson, find Anticipating misuse.	Everyone is in and on station 1.
21-43	Online: work the stations in order	Pupils read each wall, write the key fact and the challenge, then answer at the badge.	Read each wall, write the key fact and the challenge, then answer at the badge.	The worksheet has writing on it for each of the 6 stations before the badge is tapped.
43-48	Check understanding	The hinge question, hands down.	Answer the hinge question, hands down.	The split across the options.
48-56	Apply	Exam practice on the worksheet, in silence.	Exam practice, in silence.	worksheet, Exam practice a) to c)
56-60	Exit and set up the next lesson	The exit question and the retrieval link to next lesson.	Write the exit answer.	One sentence that answers the big question.

The explanation and the worked example

This is what the walls of the experience say, station by station. Teach the first two or three from the front before anyone puts a headset on; the rest the pupils read for themselves.

1. Anticipating misuse

- Thinking ahead about how users might accidentally or deliberately misuse a program.
- E.g. typing text where a number is expected, or clicking a button many times.
- Plan for every eventuality.

Challenge on this wall: List three ways a user might misuse a program that asks for their age.

2. When things go wrong

- Even with valid input, a program can crash: dividing by zero, a lost connection, a printer out of paper, or a full disk.
- Programmers use exception handling so the program copes instead of crashing.

Challenge on this wall: Fill in the four problems shown on the wall.

3. Authentication

- Checking a user is who they say they are before they can use a system.
- Usually a username and password.
- Strong passwords are long and mix upper and lower case, numbers and symbols.

Challenge on this wall: Explain the difference between validation and authentication.

4. Forgotten passwords and new users

- A reset link or code can be sent to a registered email address or phone.
- New users often confirm their email address before the account is activated.
- Two-factor authentication adds a code sent to your phone.

Challenge on this wall: Why is a reset link sent to your email, not shown on screen?

5. Bots and CAPTCHA

- Bots are programs that fill in forms automatically, e.g. to make fake accounts or guess passwords.
- CAPTCHA tests like 'I'm not a robot' are easy for humans but hard for bots.
- Audio CAPCHAs help users who can't see the images.

Challenge on this wall: Why would an attacker want a program that fills in forms automatically? Suggest two ways a website can stop one.

6. Validate a password

- It must be at least 10 characters long.
- It must contain a digit and a letter.
- It must not be one of the 100 most common passwords.

Challenge on this wall: Write the first three lines of a password validation program.

Retrieval and the check for understanding

Starter — Think of three ways someone could get into a school computer account without ever breaking the software. How could the school stop each one?

Answer space: 3 lines on the worksheet.

The answer is in the teacher answers for this unit, not on this plan: a plan left on a desk should not give it away.

Hinge question

Use this one, hands down, before anyone moves on to independent work. It is a real activity from the experience (rp-102-s4-c1), so the wording a pupil sees matches the wording you say.

Why does a website email a link to reset your password?

Correct: Only the real owner should be able to access that email account

What a pupil might choose instead

- Emails are faster than web pages
- It saves storage
- To show you adverts

Count the hands on each. If more than a few of the class are on one wrong option, that option is the misconception to teach against before anyone goes on: go back to the station this question sits on and work through its example again, aloud, before the independent work.

The course's own response to a wrong answer here: It proves you own the account before letting you change the password.

One response, not four. one response covers every wrong option; it does not say what to revisit for each one, so decide your own reteach for each option before the lesson - the list above is the start of it.

Guided to independent practice

STAGE	ACTIVITIES	WHAT IT IS
Guided	13	Support is in the activity: an example to run, an output to predict, one thing to change, a gap to fill.
Independent	0	The pupil writes or fixes it with the brief and nothing else.
Challenge	0	The hardest activity on each station. It is not optional and it is not extra: it is the last step of the station.

Every activity in this lesson is guided: a pupil chooses between options and never produces anything of their own. Plan the independent work yourself - the exam practice on the worksheet is the obvious place - and do not read a full score here as evidence that a pupil can do it unaided.

Scaffolds already in the product, and how they fade

- **Hint** is a ladder, not a door: the first rung names the idea, the next shows the shape of the code, the next how it starts, the last animates the technique on different data. A pupil asks for each rung. None of them is the answer.
- **Run** is free and unmarked; **Check my answer** marks. A pupil may run as often as they like, which is what makes trial and error safe.
- **I need help** says to ask you, and says plainly that the question still has to be completed. It does not notify anyone and it does not unlock anything.
- The worked example appears on every early activity and on none of the late ones. That fade is deliberate; it is not a missing example.

Accessibility

- **Read aloud** on every task window reads the task and its brief, with the prescribed values read as values rather than as punctuation.
- Text is 16px or larger throughout, and a question window scrolls rather than shrinking its type.
- Prescribed data is boxed and in a monospace face as well as coloured, so it still reads as code in black and white, in a high-contrast mode, and to a pupil who does not separate green from amber.
- **Lower literacy and EAL:** pre-teach the vocabulary above; let the pupil say the key fact before writing it; accept the key fact in two or three words.
- **SEND:** the sequence is compulsory but the pace is not. A pupil may finish a station next lesson; the progress saves after every answer and reopens at their earliest unfinished question.
- A headset is never required. Every experience works on a laptop or a tablet with a mouse or a finger.

Delivering the online experience

Before anyone enters

- Worksheet out, pen in hand. The rule is write first, then answer at the badge: the first answer is the one that counts.
- Say where the stations are. They are numbered around the room: 1 and 2 to the right, 3 and 4 behind, 5 and 6 to the left, and the final challenge on the floor.
- Say that a badge marked **Not yet** is not a fault. It opens when the station they are on is finished.

The stations, and what to record at each

#	STATION	ACTIVITIES	RECORD ON PAPER
1	Anticipating misuse	2	Anticipating misuse means...
2	When things go wrong	2	Problems that could crash a running program include...
3	Authentication	2	Authentication means...
4	Forgotten passwords and new users	2	If a user forgets their password...
5	Bots and CAPTCHA	2	CAPTCHA is used to...

#	STATION	ACTIVITIES	RECORD ON PAPER
6	Validate a password	2	A password is accepted if...
final	Final challenge: defend the program	1	the answer to the station's challenge

Where to pause

Stop the class after station 2 and take answers to its challenge out loud. That is the point where a pupil who has misread the technique can still be put right cheaply.

Afterwards

Headsets off and down. Ask three pupils for the key fact from a station of your choosing, in their own words and not read off the sheet. Then the exit question.

Assessment, evidence and unfinished work

ID	QUESTION	MARKS	WHERE
rp-102-x1	Describe what is meant by authentication, using an example.	2	worksheet, Exam practice a); deck, Exam practice slide
rp-102-x2	Describe two techniques a website can use to stop automated programs creating accounts.	4	worksheet, Exam practice b); deck, Exam practice slide
rp-102-x3	A program divides one number by another entered by the user. Explain how the programmer should handle a division by zero.	3	worksheet, Exam practice c); deck, Exam practice slide

Expected answers

Model answers and marking guidance are teacher-only and are not printed on this plan. They are in the unit's teacher guide pack, and the per-activity marking for this lesson is in `answers/records/rp-102.json`, which is not published with the website.

Common errors

The wrong options the activities offer are the errors this lesson expects. The commonest to watch for:

- **What does anticipating misuse mean?** — Blocking all users; Only testing with normal data
- **A user types their name into a box asking for a number. What should a** — Crash; Accept it anyway
- **How can a program cope with a problem like a missing file?** — Ignoring it; Using a longer variable name
- **What is authentication?** — Checking data is sensible; Testing a program
- **Which is the strongest password?** — password; 123456

Unfinished work

Record where each pupil stopped, not a tick. They resume at their earliest unfinished question, and a station left half-done is the thing to open next lesson. Do not mark the lesson complete for a pupil who did not finish it: the score screen already shows what was and was not done, and that is the honest record.

Exit, next lesson and homework

Exit check

One of these, written, not a show of hands:

- What defensive design considerations are
- Problems that can occur when a program runs
- Authentication techniques

The confidence columns on the worksheet are a supplement to this, not a substitute: a pupil who rates themselves confident and cannot answer the exit question has told you nothing.

Retrieval link to next lesson

Next is rp-103, Code clinic. Use this lesson's big question as next lesson's starter, so the first thing they do is retrieve it.

Homework

Finish any station left open, on the website, from the question they stopped at. If the class has no devices at home, set the exam practice on the worksheet instead and mark it next lesson.

Shorter lessons, longer lessons and no internet

A 40 to 45 minute lesson

Keep the starter, the explanation and the model at full length: cutting those is what makes the online work fail. Take the time out of the online block and the apply phase. Pupils stop where they are; their progress saves after every answer and reopens at their earliest unfinished question, so the next lesson starts there.

If the internet or the devices are unavailable

Teach from the slides and the worksheet. The slides carry the same teaching as the station walls, and the worksheet carries the key facts, the challenges and the exam practice. Set the online stations when the devices are back.

There is no offline version of the experience, and this plan does not contain one. Do not hand out a printed walkthrough of the stations as a substitute: the experiences are used on the website.

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